

Test Training 1: COLLISION-1.46M Phase 5

Quick Statistics

Dataset Version:	collision_dataset_v3
Total Tokens:	2,411,502 (Train: 2,108,753, Val: 302,749)
Model Parameters:	1,462,464 (3 layers, 128 d_model, 4 heads)
Vocabulary Capacity:	8,000 capacity (890 active tokens)
Steps Trained:	2,000 steps on CPU

Learning Metrics

Step	Train Loss	Val Loss	Perplexity	Learning Rate	Time (s)
0 (Initial)	-	5.1501	172.45	0.000000	0.0
500	4.4737	5.1501	172.45	0.000375	199.2
1000	2.8216	4.3478	77.31	0.000599	409.8
1500	2.0934	4.1409	62.86	0.000588	612.9
2000 (Final)	1.4061	4.3394	76.66	0.000564	817.8

Note: The best validation loss was 4.1409 achieved at step 1500 (perplexity: 62.86).

Generation Progression

Prompt: "Artificial intelligence"

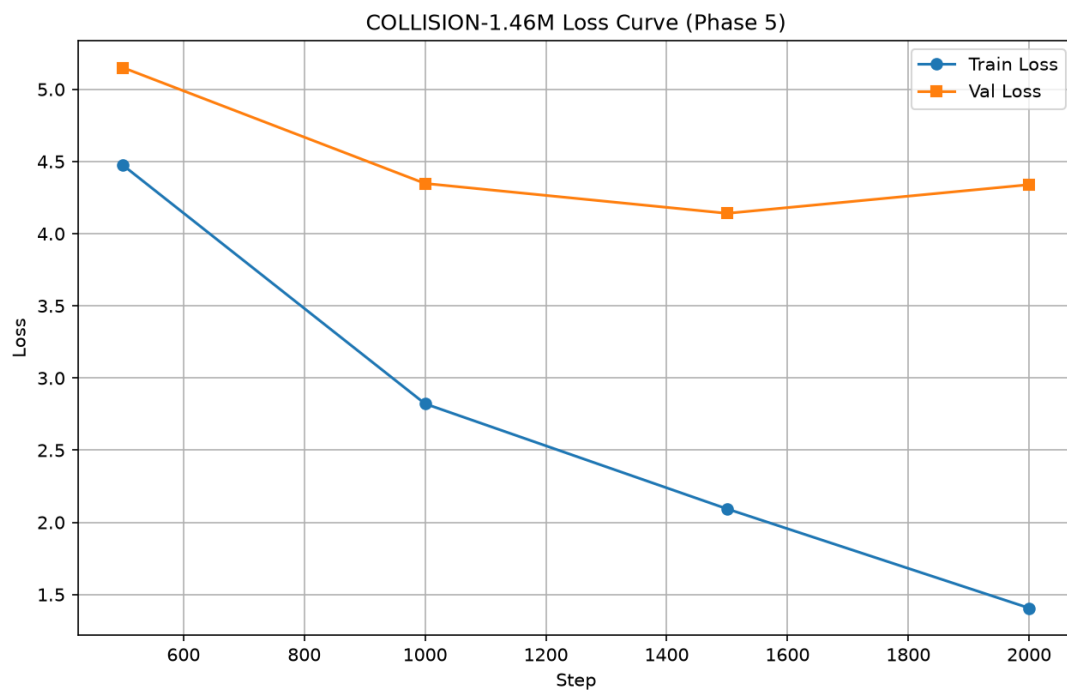
Baseline (Step 0): "Artificial intelligencelabSus "

Step 500: "Artificial intelligence a syeto ti i.isw er ,e a ofue esof . . .to h hbo ymc."

Step 1000: "Artificial intelligence daily of Section this applications of Section ddele rimagnetifergy the ced of celject m ofs of"

Step 2000: "Artificial intelligence claims. Document Section provide Section supporting methodologies. Section 7: Darra An of Experts field. Recent The Elation Ke b"

Loss Curve



Training Classification

HEALTHY

Both training and validation loss decreased substantially over the 2,000 steps compared to baseline. A slight upward trend in validation loss between steps 1500 and 2000 suggests the onset of overfitting, meaning that 1,500 steps is the optimal training length for this model/dataset configuration.